Simple lag-one model: sensitivity to transport matrix and weights

Inverse-square Exponential Lowest-cost only

Cumulative 0–50 miles

Mode-adjusted cost
Cost: rail rate ×0.5
Cost: rail rate ×2
Road distance
Straight-line distance

−0.02 −0.01 0.00 0.01 0.02 0.03 0.04 0.05

Change in soybean share (percentage points)

Cumulative 0–75 miles

Mode-adjusted cost
Cost: rail rate ×0.5
Cost: rail rate ×2
Road distance
Straight-line distance

−0.02 −0.01 0.00 0.01 0.02 0.03 0.04 0.05

Change in soybean share (percentage points)

Cumulative 0–50 miles

Mode-adjusted cost
Cost: rail rate ×0.5
Cost: rail rate ×2
Road distance
Straight-line distance

−150 −100 −50 0 50 100 150

Direct acreage estimate (thousand acres)

Cumulative 0–75 miles

Mode-adjusted cost
Cost: rail rate ×0.5
Cost: rail rate ×2
Road distance
Straight-line distance

−150 −100 −50 0 50 100 150

Direct acreage estimate (thousand acres)

100 million gal/year added capacity, averaged equally over 16 refinery locations. 95% crusher-clustered intervals.
Same 2015–2025 sample; crusher–ring and ring–year fixed effects. Non-water denominator; exploratory estimates, no multiple-testing adjustment.